

Sharp sparse Rademacher comparison for orthogonal monochromatic blocks

Candidate partial result for `fa_banach_001`

11 August 2026

Status. Candidate substantial partial result; likely valid. This is not a solution of the full arbitrary-matrix or circulant problem.

1 Source question

For a scalar matrix $A = (a_{ij})$ and $p \geq 1$, Latała and Świątkowski define

$$R_p(A) := \sup_{\|s\|_2, \|t\|_2 \leq 1} \left\| \sum_{i,j} a_{ij} \varepsilon_{ij} s_i t_j \right\|_p$$

and prove in Proposition 1.5 that

$$\frac{1}{2} M_p(A) \leq R_p(A) \lesssim \sqrt{\log(p+1)} M_p(A), \quad M_p(A) := \max_{|I| \leq p} \left\| (|a_{ij}| \mathbf{1}_{(i,j) \in I}) \right\|_{\text{op}}.$$

They then ask whether the logarithmic factor is necessary [1, Proposition 1.5 and the following sentence, p. 4].

The same paper proves the constant-factor estimate

$$R_p(B) \leq C_0 M_p(B) \tag{1}$$

for every 0–1 matrix B , with a universal constant C_0 [1, Proposition 1.4]. The result below shows that this sharp class has a dimension-free direct-sum closure that permits arbitrarily many different coefficient scales.

2 Proof intuition

An orthogonal block decomposition splits both test vectors into square-summable pieces. The source theorem controls the Rademacher sum inside each block by the global sparse norm times the product of the two corresponding block lengths. Minkowski adds these products, while Cauchy–Schwarz bounds their sum by one. Thus neither the number of blocks nor their coefficient scales can create the logarithmic accumulation seen in a generic dyadic decomposition. The unresolved case is exactly where different scales share rows or columns, destroying this two-sided square summability.

3 Direct-sum stability

We allow rectangular matrices in the statement; they can equivalently be padded by zero rows or columns.

Theorem 1 (orthogonal direct-sum stability). *Fix $p \geq 1$. Suppose A is, after independent permutations of its rows and columns, an orthogonal direct sum*

$$A = A_1 \oplus \cdots \oplus A_m.$$

If a constant C satisfies

$$R_p(A_\ell) \leq CM_p(A_\ell) \quad (1 \leq \ell \leq m),$$

then the same constant satisfies

$$R_p(A) \leq CM_p(A).$$

Proof. Fix s, t with $\|s\|_2, \|t\|_2 \leq 1$. Let s_ℓ, t_ℓ be their restrictions to the row and column spaces of A_ℓ , and put

$$\alpha_\ell = \|s_\ell\|_2, \quad \beta_\ell = \|t_\ell\|_2.$$

When these quantities are nonzero, write $s_\ell = \alpha_\ell u_\ell$ and $t_\ell = \beta_\ell v_\ell$ with u_ℓ, v_ℓ unit vectors. A zero restriction simply contributes zero. Minkowski's inequality and the assumed block estimates give

$$\begin{aligned} \left\| \sum_{i,j} a_{ij} \varepsilon_{ij} s_i t_j \right\|_p &\leq \sum_{\ell=1}^m \alpha_\ell \beta_\ell \left\| \sum_{(i,j) \text{ in block } \ell} (A_\ell)_{ij} \varepsilon_{ij} (u_\ell)_i (v_\ell)_j \right\|_p \\ &\leq C \sum_{\ell=1}^m \alpha_\ell \beta_\ell M_p(A_\ell). \end{aligned}$$

Every set of at most p entries in A_ℓ is also an admissible set in A , and embedding the resulting sparse matrix does not change its operator norm. Hence $M_p(A_\ell) \leq M_p(A)$. Since the row blocks and column blocks are respectively disjoint,

$$\sum_{\ell} \alpha_\ell^2 \leq \|s\|_2^2 \leq 1, \quad \sum_{\ell} \beta_\ell^2 \leq \|t\|_2^2 \leq 1.$$

Cauchy–Schwarz now yields

$$\sum_{\ell} \alpha_\ell \beta_\ell \leq 1,$$

so the displayed Rademacher norm is at most $CM_p(A)$. Taking the supremum over s, t proves the assertion. $\square$

Corollary 2 (orthogonal monochromatic blocks). *Let R_ℓ be pairwise disjoint row sets and C_ℓ pairwise disjoint column sets. For each ℓ , let $E_\ell \subset R_\ell \times C_\ell$ and let λ_ℓ be a scalar. Suppose that*

$$a_{ij} = \lambda_\ell \mathbf{1}_{E_\ell}(i, j) \quad \text{on } R_\ell \times C_\ell, \quad a_{ij} = 0 \quad \text{otherwise.}$$

Then, with universal constants independent of all the data,

$$R_p(A) \asymp M_p(A).$$

In particular, the factor $\sqrt{\log(p+1)}$ is unnecessary on this class, even when the number of blocks and the dynamic range of $|\lambda_\ell|$ are arbitrary.

Proof. The law defining R_p is unchanged when deterministic entry signs are absorbed into the independent Rademacher variables. Thus each nonzero block may be written as $|\lambda_\ell|B_\ell$ with B_ℓ a 0–1 matrix. Homogeneity and (1) give

$$R_p(|\lambda_\ell|B_\ell) \leq C_0 M_p(|\lambda_\ell|B_\ell).$$

Theorem 1 supplies the upper estimate for A .

For completeness, the reverse estimate is the general lower bound from the source. Indeed, choose I , $|I| \leq p$, and nonnegative unit vectors u, v that norm the nonnegative sparse matrix $(|a_{ij}|\mathbf{1}_{(i,j) \in I})$. Conditional expectation onto the signs in I shows that deleting all other summands cannot increase the L_p norm. On the event that the signs in I align with the deterministic coefficient signs, which has probability $2^{-|I|}$, the selected sum equals $M_p(A)$. Therefore its L_p norm is at least $2^{-|I|/p} M_p(A) \geq \frac{1}{2} M_p(A)$. Taking the optimal I completes the proof. $\square$

4 Verification, scope, and novelty status

The proof has no computational dependency. The key normalization identities, the local-to-global inequality $M_p(A_\ell) \leq M_p(A)$, the handling of zero blocks and signs, and the noninteger- p lower-bound constant were separately audited in `verification_report.md`.

The result does not cover a decomposition whose pieces merely have disjoint entries: disjointness is required simultaneously in their row spaces and in their column spaces. In particular, the permutation diagonals of a circulant matrix overlap in every row and column, so this theorem does not settle the main circulant conjecture. The obstruction persists in current work: arXiv:2405.13656 improves the general and circulant estimates but explicitly states that a simple sharp estimate for $R_A(p)$ remains unclear [2]; arXiv:2512.23673 still treats the broader conjecture as open [3].

The run indexes and these two later arXiv sources were searched for the exact sparse comparison, block-diagonal variants, and direct-sum terminology. No exact statement of Theorem 1 or Corollary 2 was found. Nonetheless the derivation is elementary from Proposition 1.4 of the source, so novelty confidence is modest. The recommended human review is to verify the scope and decide whether this closure lemma is useful enough to retain as a substantial partial result.

References

- [1] R. Latała and W. Świątkowski, *Norms of Randomized Circulant Matrices*, arXiv:2106.03139, <https://arxiv.org/abs/2106.03139>.
- [2] R. Latała, *On the spectral norm of Rademacher matrices*, arXiv:2405.13656, <https://arxiv.org/abs/2405.13656>.
- [3] R. Meller, *Spectral norm of matrices with independent entries up to polyloglog*, arXiv:2512.23673, <https://arxiv.org/abs/2512.23673>.

In the general case, a similar result holds. However, here the upper and lower bounds are not of the same order.

Proposition 1.5. *For any matrix $A = (a_{ij})_{i,j \leq n}$ and $p \geq 1$,*

$$\frac{1}{2} \max_{I \subset [n] \times [n], |I| \leq p} \|(|a_{ij}| \mathbb{1}_{(i,j) \in I})\| \leq \|A\|_{\varepsilon, p} \lesssim \sqrt{\ln(p+1)} \max_{I \subset [n] \times [n], |I| \leq p} \|(|a_{ij}| \mathbb{1}_{(i,j) \in I})\|.$$

We do not know whether the logarithmic factor is necessary.

Organization of the paper. In Section 2 we prove the main results of the paper, i.e. Theorems 1.1 and 1.3. In Section 3 we study norms $\|A\|_{\varepsilon, p}$ and establish Propositions 1.4 and 1.5. We also provide there estimates for $\|A\|_{\varepsilon, p}$ in the case when A are adjacency matrices of hypercubes $\mathbb{Z}_2^d$ and more general discrete tori $\mathbb{Z}_m^d$. The adjacency matrix of $\mathbb{Z}_m^d$ is not circulant, however it is very close to have such a property and Corollary 3.5 shows that we may apply previously derived estimates to get two-sided bounds on $\mathbb{E} \|(\mathbb{1}_{i \sim j} \varepsilon_{ij})_{i,j \in \mathbb{Z}_m^d}\|$. In the last section of the paper we extend Theorem 1.1 and Conjecture 1.2 to the case of random matrices with independent entries X_{ij} satisfying the mild regularity condition $\|X_{ij}\|_{2p} \leq \alpha \|X_{ij}\|_p$ for $p \geq 2$. We show that results of [8] imply that more general Conjecture 4.3 holds in the case when X_{ij} are mixtures of Gaussian variables. We also show that to establish formulated conjectures it is enough to prove a slightly weaker bound (28).

2 Proofs of main results

In the sequel we will frequently compare L_p -norms of real and vector-valued Rademacher sums $S = \sum_{k=1}^n x_k \varepsilon_k$, where x_i are vectors from a normed space $(F, \|\cdot\|)$ and ε_i are independent symmetric ± 1 r.v.'s. The classical Khintchine (in the case when coefficients x_i are real) and the Kahane-Khintchine inequalities, cf. [9, Section 4.3], state that for $p > q > 0$,

$$\|S\|_q \leq \|S\|_p \leq C_{p,q} \|S\|_q,$$

where $\|S\|_q = (\mathbb{E} \|S\|^q)^{1/q}$ and $C_{p,q}$ is a constant depending only on p and q . Moreover for $p > q > 1$, $C_{p,q} \leq \sqrt{\frac{p-1}{q-1}}$, therefore for $p \geq 2$, $C_{p,2} \leq \sqrt{p}$ and for $p \geq 1 + (e-2)^{-1}$, $C_{2p,p} \leq \sqrt{e}$.

Markov's inequality yields $\mathbb{P}(\|S\| \geq e \|S\|_p) \leq e^{-p}$. Using the Paley-Zygmund inequality

$$\mathbb{P}\left(Z \geq \frac{1}{2} \mathbb{E} Z\right) \geq \frac{1}{4} \frac{(\mathbb{E} Z)^2}{\mathbb{E} Z^2} \quad \text{for nonnegative r.v. } Z,$$

one may derive a reverse lower tail bound for $\|S\|$ (similar estimates can be found in [3, 7], we present details for the sake of completeness). For any $p \geq 1 + (e-2)^{-1}$, by taking $Z = \|S\|^p$ we obtain

$$\mathbb{P}\left(\|S\| \geq \frac{1}{2} \|S\|_p\right) \geq \frac{1}{4} \left(\frac{\|S\|_p}{\|S\|_{2p}}\right)^{2p} \geq \frac{1}{4} e^{-p}. \quad (5)$$

Figure 1: The source question on arXiv PDF page 4.